ADVANCED EXCEL · LESSON 7 OF 12

Lookup Functions: VLOOKUP, XLOOKUP and INDEX-MATCH

Bring data together from different tables. Master VLOOKUP and HLOOKUP (exact and approximate match), the modern XLOOKUP, the flexible INDEX + MATCH combination (including two-way and reverse lookups), CHOOSE, and how to troubleshoot

● Live classes 4–5 ✦ Excel for Microsoft 365, 2026 Intermediate ⏱ 120 min

By the end of this lesson you can

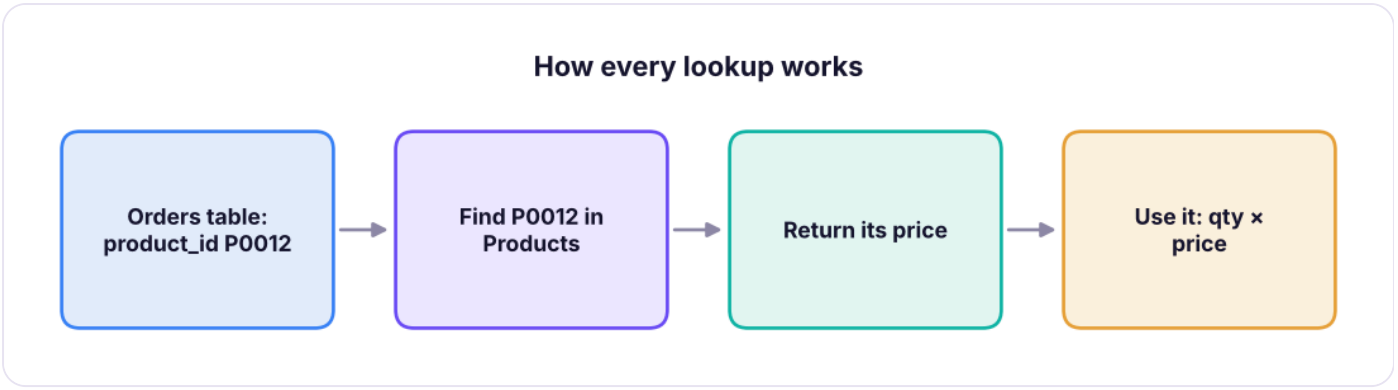
- Explain why lookups are the Excel equivalent of a SQL join
- Use VLOOKUP and HLOOKUP with exact and approximate match, and know their limits
- Use XLOOKUP for left lookups, default values, last-match search and multiple return columns
- Combine INDEX and MATCH for reverse (left) and two-way (matrix) lookups
- Use CHOOSE, and fix the common causes of

Datasets used

`products.csv` `superstore_sales.csv`

Why lookups matter

Business data is split across tables: orders in one, products in another, customers in a third. A **lookup** fetches a value from one table using a key that both tables share, such as a product ID or employee ID.



A lookup takes a key from one table, finds it in another table, and returns a value from the same row.

> **TIP** **You are already learning SQL joins**
>
> A lookup is a **left join** for one column at a time: keep every row of the main table and bring in matching information from the other table. You will see the same idea as `JOIN` in SQL (SQL Lesson 8), `merge()` in pandas, and relationships in Power BI.

We will use this lookup table on a sheet called `Prod`:

Prod: product lookup table (A1:E7)

	A	B	C	D	E
1	**Product ID**	**Product**	**Category**	**Price**	**Stock**
2	P101	Office Chair	Furniture	4500	32
3	P102	Standing Desk	Furniture	18500	8
4	P103	Laser Printer	Technology	12900	15
5	P104	Notebook Pack	Office Supplies	240	540
6	P105	Smartphone	Technology	21999	27
7	P106	Bookcase	Furniture	7600	0

VLOOKUP

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

- **lookup_value**: what you are looking for (the key).
- **table_array**: the table to search; the key **must be in its first column**.
- **col_index_num**: which column of that table to return (1 = first column).
- **range_lookup**: FALSE (or 0) for an **exact** match. Use TRUE only for approximate matching on sorted bands.

Order lines looking up product details with VLOOKUP

	A	B	C	D	E	F
1	Order	Product ID	Qty	Product =VLOOKUP(B2,Prod!$A$2:$E$7,2,FALSE)	Price	Amount
2	1001	P103	2	Laser Printer	12,900	25,800
3	1002	P101	6	Office Chair	4,500	27,000
4	1003	P105	1	Smartphone	21,999	21,999
5	1004	P104	20	Notebook Pack	240	4,800
6	1005	P109	3	#N/A	#N/A	#N/A

D2 `=VLOOKUP(B2,Prod!$A$2:$E$7,2,FALSE)`

F2 `=C2*E2`

Each pattern is copied down/across to the other cells in its column.

Row 6 shows `#N/A` because P109 does not exist in the product table. That is the correct answer: the key is not there.

COMMON MISTAKES

The four classic VLOOKUP mistakes

1. **Forgetting FALSE**: the default is approximate match, which silently returns a *wrong* row for unsorted data.
2. **Not locking the table**: `A2:E7` copied down becomes `A3:E8`, so the last rows are missed. Use `$A$2:$E$7` or a Table name.
3. **Hard-coded column numbers**: inserting a column inside the table changes which column 4 means. Use `MATCH` for the column number, or XLOOKUP.
4. **Key not in the first column**: VLOOKUP cannot look to the left (use INDEX + MATCH or XLOOKUP).

Approximate match: bands and slabs

With `TRUE`, VLOOKUP finds the **largest value less than or equal to** the lookup value. The first column must be sorted ascending. This is perfect for tax slabs, commission tiers, discount bands and grades.

Approximate match for commission tiers

	A	B	C	D	E	F	G
1	**Sales from**	**Commission**		**Rep**	**Sales**	**Rate**	**Commission**
2	0	0%		Asha	85,000	0%	0
3	100000	3%		Ravi	240,000	3%	7,200
4	250000	5%		Meera	250,000	5%	12,500
5	500000	7%		Kabir	610,000	7%	42,700
6	1000000	10%		Zoya	1,200,000	10%	120,000

F2 `=VLOOKUP(E2,$A$2:$B$6,2,TRUE)`

G2 `=E2*F2`

Each pattern is copied down/across to the other cells in its column.

Ravi's 2,40,000 falls between 1,00,000 and 2,50,000, so he gets the 1,00,000 row: 3%. Meera's 2,50,000 matches the 2,50,000 row exactly: 5%.

HLOOKUP

HLOOKUP is VLOOKUP turned sideways: it searches the **first row** and returns a value from a row below. Use it when data runs across columns (for example months as headers).

HLOOKUP across a horizontal table

	A	B	C	D	E	F	G	H
1	**Month**	**Jan**	**Feb**	**Mar**	**Apr**		**Find month**	Mar
2	Target	120	130	145	150		Target →	145
3	Actual	118	136	139	162		Actual →	139

H2 `=HLOOKUP(H1,$A$1:$E$3,2,FALSE)`

H3 `=HLOOKUP(H1,$A$1:$E$3,3,FALSE)`

XLOOKUP: the modern lookup (Microsoft 365 / Excel 2021+)

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found],
[match_mode], [search_mode])
```

XLOOKUP fixes every VLOOKUP limitation:

Feature	VLOOKUP	XLOOKUP
Default match	Approximate (risky)	**Exact**
Look to the left	No	**Yes**: lookup and return arrays are separate
Column inserted in the table	Breaks the index number	**Safe**
Custom "not found" message	Needs IFERROR	Built-in 4th argument
Return several columns	One at a time	**Spills** many columns
Search last to first	No	`search_mode = -1`
Wildcards	Yes	`match_mode = 2`
Next smaller / larger match	Sorted data only	`match_mode = -1` or 1, any order

XLOOKUP examples (lookup table on sheet Prod)

	A	B	C
1	**Need**	**Formula**	**Result**
2	Price of P105	=XLOOKUP("P105",Prod!A2:A7,Prod!D2:D7)	21,999
3	ID of "Bookcase" (left lookup)	=XLOOKUP("Bookcase",Prod!B2:B7,Prod!A2:A7)	P106
4	Missing ID with a message	=XLOOKUP("P109",Prod!A2:A7,Prod!B2:B7,"Not found")	Not found
5	Product, category and price at once	=XLOOKUP("P103",Prod!A2:A7,Prod!B2:D7)	Laser Printer
6	Last Furniture item in the list	=XLOOKUP("Furniture",Prod!C2:C7,Prod!B2:B7,,0,-1)	Bookcase
7	Product name containing "Desk"	=XLOOKUP("*Desk*",Prod!B2:B7,Prod!D2:D7,,2)	18,500

```
C2   =XLOOKUP("P105",Prod!A2:A7,Prod!D2:D7)
C3   =XLOOKUP("Bookcase",Prod!B2:B7,Prod!A2:A7)
C4   =XLOOKUP("P109",Prod!A2:A7,Prod!B2:B7,"Not found")
C5   =XLOOKUP("P103",Prod!A2:A7,Prod!B2:D7)
C6   =XLOOKUP("Furniture",Prod!C2:C7,Prod!B2:B7,,0,-1)
C7   =XLOOKUP("*Desk*",Prod!B2:B7,Prod!D2:D7,,2)
```

The fourth row returns three columns at once: the result spills across C5:E5.

XLOOKUP for bands (no sorting needed)

`match_mode = -1` returns the exact match or the **next smaller** item, like VLOOKUP's TRUE, but the table does not need to be sorted:

`=XLOOKUP(E2, $A$2:$A$6, $B$2:$B$6, , -1)` gives the same commission rates as the approximate VLOOKUP above.

> **TIP** **XMATCH**
>
> `XMATCH` is the modern version of MATCH with the same match and search modes as XLOOKUP. `=XMATCH("P105",Prod!A2:A7)` → 5.

INDEX and MATCH

Before XLOOKUP, professionals used **INDEX + MATCH**, and you will still see it everywhere. It works in every Excel version and is very flexible.

- `=INDEX(array, row_num, [col_num])` returns the value **at a position**.
- `=MATCH(lookup_value, lookup_array, [match_type])` returns the **position** of a value. Use `0` for exact match.

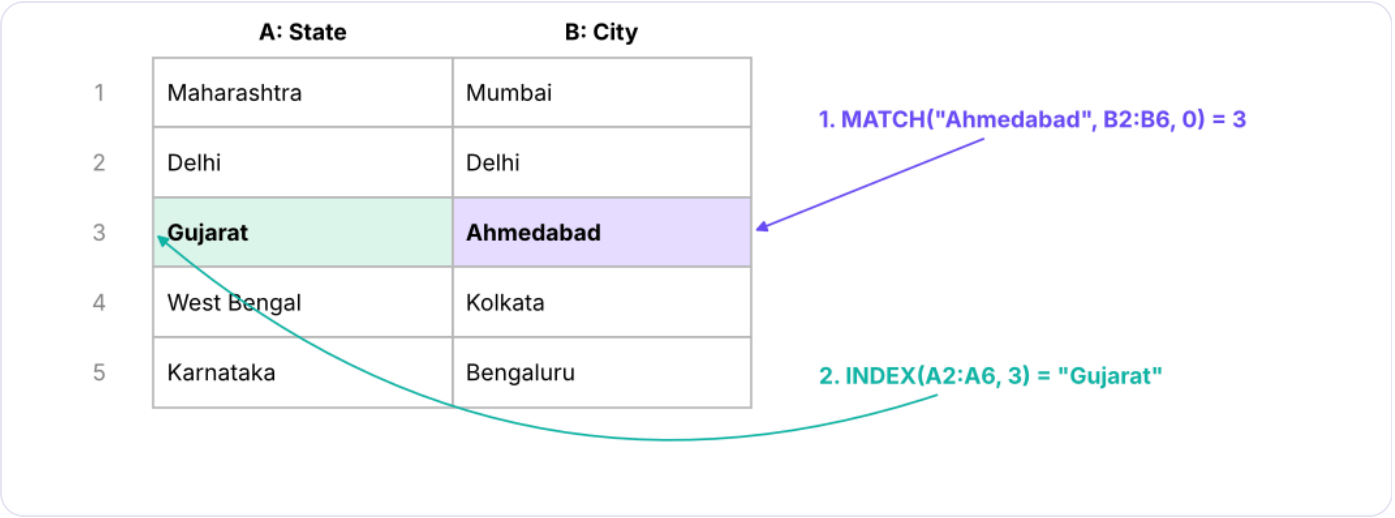
X INDEX returns a value at a position; MATCH returns the position of a value

	A	B	C	D	E
1	**State**	**City**	**Population 2011 (mn)**		**Formula**
2	Maharashtra	Mumbai	18.4		=INDEX(A2:C6,2,1)
3	Delhi	Delhi	16.3		=MATCH("Ahmedabad",B2:B6,0)
4	Gujarat	Ahmedabad	6.4		=MATCH("Population 2011 (mn)",A1:C1,0)
5	West Bengal	Kolkata	14.1		=INDEX(C2:C6,MATCH("Kolkata",B2:B6,0))
6	Karnataka	Bengaluru	8.5		=INDEX(A2:A6,MATCH("Ahmedabad",B2:B6,0))

```
F2   =INDEX(A2:C6,2,1)
F3   =MATCH("Ahmedabad",B2:B6,0)
F4   =MATCH("Population 2011 (mn)",A1:C1,0)
F5   =INDEX(C2:C6,MATCH("Kolkata",B2:B6,0))
F6   =INDEX(A2:A6,MATCH("Ahmedabad",B2:B6,0))
```

Reverse (left) lookup

In the table above, City is in column B and State is in column A. VLOOKUP cannot return a column to the **left** of the key, but INDEX + MATCH can: `=INDEX(A2:A6, MATCH("Ahmedabad", B2:B6, 0))` → **Gujarat**.



INDEX + MATCH: MATCH finds the row number of the city; INDEX returns the state on that row.

Two-way (matrix) lookup

INDEX can take a row **and** a column number. Use one MATCH for each to look up the intersection of a row and a column, for example an employee's salary from a table where both the employee and the field are chosen in drop-downs.

Two-way lookup: =INDEX($B$2:$E$6, MATCH($G3,$A$2:$A$6,0), MATCH(H$2,$B$1:$E$1,0))

	A	B	C	D	E	F	G	H	
1	EmpName	DepartmentID	JobTitle	Salary	City				
2	Employee_1	D10	Developer	82000	Pune		Employee	JobTitle	Sal
3	Employee_2	D20	Analyst	68000	Delhi		Employee_4	Manager	1
4	Employee_3	D10	Tester	54000	Pune		Employee_2	Analyst	
5	Employee_4	D30	Manager	125000	Mumbai				
6	Employee_5	D20	Developer	79000	Delhi				

```
H3   =INDEX($B$2:$E$6,MATCH($G3,$A$2:$A$6,0),MATCH(H$2,$B$1:$E$1,0))
I3   =INDEX($B$2:$E$6,MATCH($G3,$A$2:$A$6,0),MATCH(I$2,$B$1:$E$1,0))
H4   =INDEX($B$2:$E$6,MATCH($G4,$A$2:$A$6,0),MATCH(H$2,$B$1:$E$1,0))
I4   =INDEX($B$2:$E$6,MATCH($G4,$A$2:$A$6,0),MATCH(I$2,$B$1:$E$1,0))
```

Because the headers in row 2 (JobTitle, Salary) are matched by MATCH, you can change a header to `City` and the formula returns cities without editing it. The XLOOKUP version nests one XLOOKUP inside another: `=XLOOKUP($G3,$A$2:$A$6,XLOOKUP(H$2,$B$1:$E$1,$B$2:$E$6))`.

INDEX + MATCH with multiple conditions

To match on two columns at once (for example Region **and** Product), multiply the conditions inside MATCH:

`=INDEX(F2:F9, MATCH(1, (B2:B9="North")*(C2:C9="Tablet"), 0))` → returns the revenue for the North + Tablet row. In Microsoft 365 this works as a normal formula; in older Excel press `Ctrl + Shift + Enter`. With XLOOKUP: `=XLOOKUP(1,(B2:B9="North")*(C2:C9="Tablet"),F2:F9)`.

CHOOSE, and VLOOKUP with CHOOSE

`=CHOOSE(index_num, value1, value2, …)` returns the item at a position.

CHOOSE examples

	A	B	C
1	**Input**	**Formula**	**Result**
2	2	=CHOOSE(A2,"Bronze","Silver","Gold")	Silver
3	27-04-2026	=CHOOSE(WEEKDAY(A3,2),"Mon","Tue","Wed","Thu","Fri","Sat","Sun")	Mon
4	15-08-2026	="Q"&CHOOSE(MONTH(A4),4,4,4,1,1,1,2,2,2,3,3,3)	Q2

```
C2   =CHOOSE(A2,"Bronze","Silver","Gold")
C3   =CHOOSE(WEEKDAY(A3,2),"Mon","Tue","Wed","Thu","Fri","Sat","Sun")
C4   ="Q"&CHOOSE(MONTH(A4),4,4,4,1,1,1,2,2,2,3,3,3)
```

The last example gives the **Indian financial-year quarter** (April–June = Q1), which a plain `ROUNDUP(MONTH()/3)` cannot do.

VLOOKUP + CHOOSE was the old trick for a left lookup: `CHOOSE({1,2}, B2:B6, A2:A6)` builds a virtual two-column table with City first and State second, so VLOOKUP can search it:

`=VLOOKUP("Ahmedabad", CHOOSE({1,2}, B2:B6, A2:A6), 2, FALSE)` → Gujarat.

Today, prefer XLOOKUP or INDEX + MATCH: they are easier to read and faster.

Troubleshooting #N/A

Cause	How to check	Fix
The key really is missing	Search the lookup column with `Ctrl + F`	Return a message: `IFNA(…,"Not found")` or XLOOKUP's 4th argument
Extra spaces	`=LEN(A2)` is longer than expected	`TRIM` both sides, or `XLOOKUP(TRIM(B2),TRIM(Prod!A2:A7),…)`
Number vs text (`101` vs `"101"`)	`=ISTEXT(B2)` differs from the lookup column	Convert one side: `VALUE()` or `TEXT(…,"0")`
Approximate match on unsorted data	Wrong but no error	Use exact match (FALSE / 0)
Range not locked	Errors appear only in lower rows	Add $ or use a Table
Invisible characters	`=CODE(RIGHT(B2))` returns 160 or 10	`SUBSTITUTE` and `CLEAN`

2026 UPDATE **Lookups in 2026**

- **XLOOKUP** and **XMATCH** are now the standard; use VLOOKUP only when sharing files with very old Excel versions.
- **FILTER** returns **all** matching rows, not just the first (Lesson 6).
- For joining large tables regularly, **Power Query → Merge Queries** (Lesson 12) is faster and repeatable, and it is exactly a SQL join.

Case: pricing the Superstore order lines

Suppose the order lines arrive without a price and you must look it up from the product master (`products.csv`). This is a lookup on 6,323 rows. pandas does the same as XLOOKUP with `merge`:

Python

```python
import pandas as pd
orders = pd.read_csv("superstore_sales.csv")[["order_id", "product_id", "quantity"]]
products = pd.read_csv("products.csv")
priced = orders.merge(products[["product_id", "product_name", "list_price"]],
on="product_id", how="left")   # = XLOOKUP for every row
priced["gross_amount"] = priced.quantity * priced.list_price
print("Rows:", len(priced), "| unmatched product IDs:", priced.list_price.isna().sum())
priced.head(5)
```

OUTPUT

```
Rows: 6323 | unmatched product IDs: 0
```

	order_id	product_id	quantity	product_name	list_price	gross_amount
0	IN-2023-00001	P0059	2	Nova Phone 223	18,000	36,000
1	IN-2023-00001	P0050	1	Orbit Appliance 798	3,850	3,850
2	IN-2023-00001	P0011	4	Orbit Table 248	15,520	62,080
3	IN-2023-00002	P0037	2	Crest Storage 946	3,260	6,520
4	IN-2023-00003	P0018	3	Crest Bookcase 170	11,710	35,130

5 rows × 6 columns

Zero unmatched IDs tells you every order line found its product: the same check you should do in Excel with `=COUNTIF(price_column,"#N/A")` or `=SUM(--ISNA(price_column))`.

Summary

KEY TAKEAWAYS

- Lookups bring columns from one table into another through a shared key, like a SQL left join.
- VLOOKUP: key in the first column, column number counted from 1, **FALSE for exact match**, lock the table with $.
- Approximate match (TRUE / match_mode −1) is for sorted bands: tax slabs, commissions, grades.
- XLOOKUP is exact by default, looks left, has a built-in "not found" value, returns several columns and can search from the end.
- INDEX returns a value at a position; MATCH finds the position. Together they do reverse and two-way lookups in any Excel version.
- Most #N/A errors come from missing keys, spaces, or number-vs-text mismatches.

Quick quiz

6 questions · tap an answer to check it

1. `=VLOOKUP("P103",A2:E7,4,FALSE)` **returns a value from which column of the range?**

A A

B C

C D

D E

Answer: C. Column index 4 counts from the first column of the table array: A=1, B=2, C=3, D=4.

2. Why does VLOOKUP return the wrong price (not #N/A) for unsorted data?

A The fourth argument was omitted, so it used approximate match

B The table was not locked with $

C The column number was too large

D VLOOKUP cannot handle text keys

Answer: A. Without FALSE, VLOOKUP uses approximate match, which assumes sorted data.

3. Which formula returns the State for the city in H2, where State is column A and City is column B?

A `=VLOOKUP(H2,A2:B6,1,FALSE)`

B `=INDEX(A2:A6,MATCH(H2,B2:B6,0))`

C `=MATCH(H2,A2:A6,0)`

D `=HLOOKUP(H2,A2:B6,1,FALSE)`

Answer: B. MATCH finds the row of the city in column B; INDEX returns column A on that row.

4. Sales of 2,40,000 with tiers at 0, 1,00,000, 2,50,000 and 5,00,000 (approximate match). Which tier is used?

A 0

B 1,00,000

C 2,50,000

D 5,00,000

Answer: B. Approximate match returns the largest tier less than or equal to the value.

5. What does XLOOKUP's 4th argument do?

A Sets the match mode

B Returns a value when nothing is found

C Searches from last to first

D Returns several columns

Answer: B. `if_not_found` replaces #N/A with your own value or message.

6. A lookup returns #N/A even though the ID is visible in both tables. Which is NOT a likely cause?

A One side has trailing spaces

B One side is a number and the other is text

C The workbook is saved as .xlsx

D There is an invisible non-breaking space

Answer: C. File format does not affect lookups. Spaces, invisible characters and type mismatches are the usual culprits.

EXERCISE

Grade lookup

Create a grade table: 0 → F, 40 → D, 55 → C, 70 → B, 85 → A. Write a formula that returns the grade for a mark in B2 using (a) VLOOKUP, (b) XLOOKUP and (c) INDEX + MATCH.

Show solution (worked answer)

With the table in E2:F6 (marks ascending): (a) `=VLOOKUP(B2,$E$2:$F$6,2,TRUE)` (b) `=XLOOKUP(B2,$E$2:$E$6,$F$2:$F$6,,-1)` (c) `=INDEX($F$2:$F$6,MATCH(B2,$E$2:$E$6,1))` (match_type 1 = largest value ≤ B2, needs ascending order).

EXERCISE

Two-way lookup

Make a table of 4 regions (rows) × 4 quarters (columns) with sales figures. Add drop-downs for region (H1) and quarter (H2) and return the matching sales value.

Show solution (worked answer)

With regions in A2:A5, quarters in B1:E1 and values in B2:E5: `=INDEX($B$2:$E$5,MATCH($H$1,$A$2:$A$5,0),MATCH($H$2,$B$1:$E$1,0))` or `=XLOOKUP($H$1,$A$2:$A$5,XLOOKUP($H$2,$B$1:$E$1,$B$2:$E$5))`. Make the drop-downs with Data Validation → List, source `$A$2:$A$5` and `$B$1:$E$1`.

INTERVIEW CORNER

Questions you may be asked

Q: What are the limitations of VLOOKUP and how do you overcome them? It cannot look left, it defaults to approximate match, it breaks when columns are inserted (hard-coded index), and it returns only the first match. Use XLOOKUP or INDEX + MATCH, and FILTER for multiple matches.

Q: Why is INDEX + MATCH considered more flexible than VLOOKUP? The lookup and return ranges are separate, so it can look left, it is not affected by inserted columns, it supports two-way lookups, and it can be faster on large sheets because it only reads the columns it needs.

Q: How would you look up a value using two criteria? `XLOOKUP(1,(range1=x)*(range2=y),return_range)`, `INDEX/MATCH` with the same multiplied condition, or a helper column that concatenates the two keys.

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